Computer Laboratory Project

Object Oriented Programming

(CED-225L)

Fall 2024

Computer Engineering Technology

Sir Syed University of Engineering & Technology
University Road, Karachi 75300
http://www.ssuet.edu.pk

LIBRARY MANAGEMENT SYSTEM

PROJECT REPORT

S.NO	Table of Contents:	PAGE NO.
1	ABSTRACT	2
2	INTRODUCTION	2
3	DEFINATION & ACRONYMS	2 - 3
4	PROPOSED SYSTEM	3
5	FEASIBILITY REPORT	4
6	TECHNICAL FEASIBILITY	5
7	TECHNICAL DESCRIPTION	5
8	GRAPHICAL USER INTERFACE (GUI)	5
9	PROJECT DESIGN DESCRIPTION	5
10	SCOPE OF THE DEVELOPMENT PROJECT	6
11	SOFTWARE REQUIRMENT SPECIFICATION (SRS)	6
12	DATA FLOW DIAGRAM (DFD)	6
13	SOURCE CODE	6 – 13
14	CONCLUSION & RECOMMENDATIONS	14
15	ACKNOWLEDGEMENTS	14
16	REFERENCES	14

1. Abstract

This project presents a Library Management System developed using Java (Swing) to automate basic library operations such as student registration, viewing books, issuing books, returning books, and generating a simple bill receipt. The system is designed for beginners and small institutions to reduce manual workload, improve accuracy, and provide a user-friendly interface.

2. Introduction

Libraries play an important role in educational institutions. Manual library systems are time-consuming and error-prone. The proposed system provides a simple computerized solution using Java GUI components to manage essential library activities efficiently.

3. Definitions and Acronyms

A. **Library Management System (LMS)**
A Library Management System is a software application used to manage basic library operations such as student registration, book issuing, book returning, and viewing book records.

B. **Graphical User Interface (GUI)**
A Graphical User Interface allows users to interact with the system using dialog boxes, menus, and buttons instead of typing commands.

C. **Java**
Java is an object-oriented programming language used to develop platform-independent applications.

D. **Array**
An array is a data structure used to store multiple values of the same data type in a single variable.

E. **Book Issue Process**
The book issue process is the activity of issuing a book to a registered student and updating the available book quantity.

F. **Book Return Process**
The book return process is the activity of returning an issued book to the library and updating book records.

G. **Bill Receipt**
A bill receipt is a summary generated by the system showing charges related to book issue or book return.

H. **Temporary Data Storage**
Temporary data storage refers to storing data in program variables and arrays, which is lost when the program is closed.

I. **Menu-Driven System**
A menu-driven system is a program that allows users to select options from a list of choices.

J. **Standalone Application**
A standalone application is a software program that runs independently on a computer without requiring internet or database connectivity.

Acronym	Full Form
LMS	Library Management System
GUI	Graphical User Interface
DFD	Data Flow Diagram
SRS	Software Requirement Specification
JDK	Java Development Kit
IDE	Integrated Development Environment
JVM	Java Virtual Machine
OOP	Object-Oriented Programming

4. Proposed System

The proposed system is a standalone Java application that:

- Registers students and creates a library card
- Displays available books
- Issues books to registered students
- Accepts returned books
- Generates a bill receipt on return

5. Feasibility Report

5.1 General Requirement Feasibility

The Library Management System is easy to use and does not require advanced technical knowledge. It can run on basic computer systems with minimal hardware and software requirements. The system is cost-effective, requires no internet or database, and is suitable for small libraries and academic projects. Maintenance and operation are simple and affordable.

Key Points:

- Easy to understand and operate by librarians and students
- Requires minimal hardware such as a basic computer or laptop
- No need for internet connection or external database
- Uses free software tools like Java JDK and NetBeans IDE
- Low development, installation, and maintenance cost
- Suitable for small libraries and academic mini-projects
- Reduces manual workload and improves work efficiency

5.2 Problems in the Current System

The current manual library system relies on paper records, which are difficult to manage and maintain. There is a high chance of data loss, duplication, and human errors. Book searching, issuing, and returning processes are time-consuming and inefficient. Maintaining accurate records becomes difficult as the number of books and students increases.

Key Problems:

- Manual record keeping using registers and files
- High chances of human errors and incorrect entries
- Risk of data loss due to misplacement or damage of records
- Time-consuming process for searching and issuing books
- Difficulty in maintaining accurate book availability records
- No quick method to generate receipts or reports
- Increased workload for librarians and staff

6. Technical Feasibility

The system is technically feasible as it uses:

- Java programming language
- Swing for GUI
- Simple arrays for data handling
  No external database or internet connection is required.

7. Technical Description

- **Programming Language**: Java
- **IDE**: NetBeans IDE 8.2
- **GUI Framework**: Java Swing
- **Data Handling**: Arrays and variables

8. Graphical User Interface (GUI)

The GUI is designed using JOptionPane dialogs. It provides menu-driven options for all operations, making it simple and interactive for users.

9. Project Design Description

A. Purpose

The purpose of this project is to provide a simple and efficient system to manage library activities and help students understand basic Java concepts.

B. Scope

- Small school or college libraries
- Academic mini-projects
- Learning Java GUI programming

10. Scope of the Development Project

The project focuses on core library functions without complex database integration.

A. Database Connectivity Tier

This version does not use database connectivity. Data is stored temporarily using arrays.

11. Software Requirement Specification (SRS)

A. Hardware Requirements

- Computer or Laptop
- Minimum 2 GB RAM
- 500 MB free disk space

B. Software Requirements

- Windows OS
- JDK 8
- NetBeans IDE 8.2

12. Data Flow Diagrams (DFD)

- **Level 0**: User interacts with LMS
- **Level 1**: Registration, Book Issue, Book Return, Receipt Generation

13. Source Code

```
import javax.swing.*;
class LIBRARYMANAGEMENT_SYSTEM {
   static boolean registered = false;
   static String name, roll, dept, email, address;
   static int semester;
   static long phone;
   static String[] bookId = {"B101", "B102", "B103", "B104", "B105"};
```

```
static String[] bookName = {
    "Java Programming",
    "Data Structures",
    "Operating System",
    "DBMS",
    "Computer Networks"
};
static int[] quantity = {3, 4, 2, 3, 2};
static int[] charge = {50, 60, 70, 65, 55};
public static void main(String[] args) {
    while (true) {
        String choice = JOptionPane.showInputDialog(
            "LIBRARY MANAGEMENT SYSTEM\n\n"
          + "1. Make Library Card\n"
          + "2. View Library Card\n"
          + "3. View Books\n"
          + "4. Issue Book\n"
          + "5. Return Book (Bill Receipt)\n"
          + "6. Exit\n\n"
          + "Enter choice:"
        );
        if (choice == null) break;
        switch (choice) {
            case "1":
                register();
                break;
            case "2":
                viewCard();
```

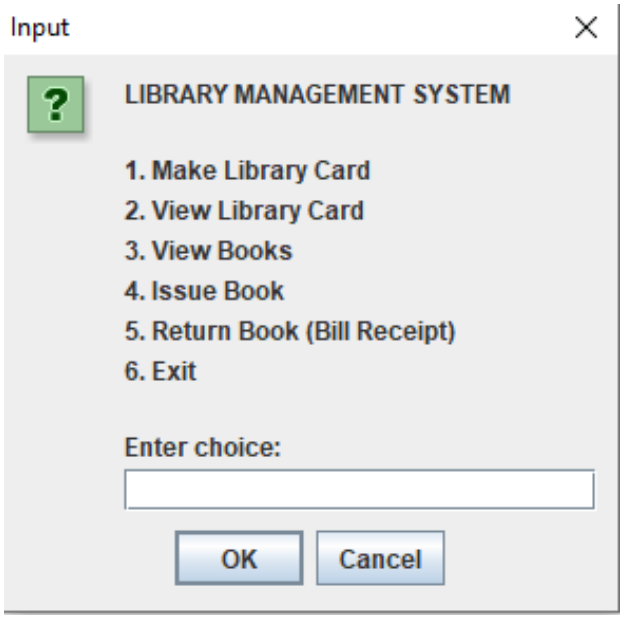
```
            break;
        case "3":
            viewBooks();
            break;
        case "4":
            issueBook();
            break;
        case "5":
            returnBook();
            break;
        case "6":
            System.exit(0);
        default:
            JOptionPane.showMessageDialog(null, "Invalid choice!");
    }
  }
}
static void register() {
  name = JOptionPane.showInputDialog("Enter Name:");
  roll = JOptionPane.showInputDialog("Enter Roll No:");
  dept = JOptionPane.showInputDialog("Enter Department:");
  semester = Integer.parseInt(JOptionPane.showInputDialog("Enter Semester:"));
  phone = Long.parseLong(JOptionPane.showInputDialog("Enter Phone Number:"));
  email = JOptionPane.showInputDialog("Enter Email:");
  address = JOptionPane.showInputDialog("Enter Address:");
  registered = true;
  JOptionPane.showMessageDialog(null, "Library Card Created Successfully!");
}
```

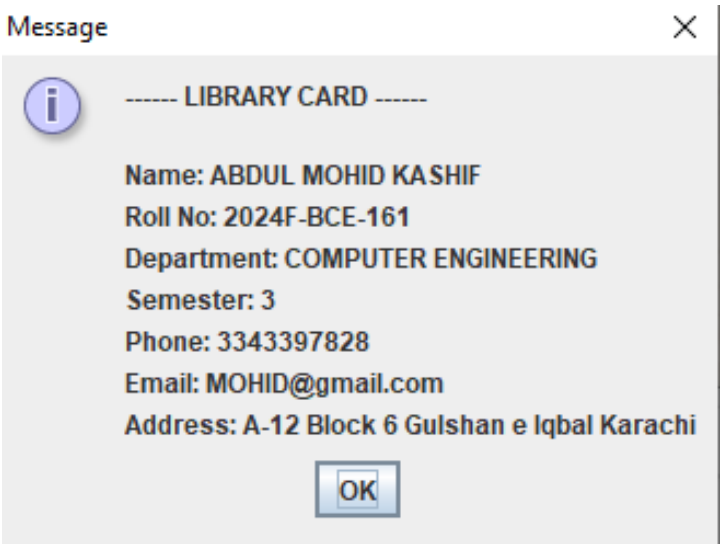
```
static void viewCard() {
    if (!registered) {
        JOptionPane.showMessageDialog(null, "No Library Card Found!");
        return;
    }
    String card =
         "------ LIBRARY CARD ------\n\n"
        + "Name: " + name + "\n"
        + "Roll No: " + roll + "\n"
        + "Department: " + dept + "\n"
        + "Semester: " + semester + "\n"
        + "Phone: " + phone + "\n"
        + "Email: " + email + "\n"
        + "Address: " + address;
    JOptionPane.showMessageDialog(null, card);
}
static void viewBooks() {
    String msg = "";
    for (int i = 0; i < bookId.length; i++) {
        msg += "ID: " + bookId[i]
            + "\nName: " + bookName[i]
            + "\nAvailable: " + quantity[i]
            + "\nIssue Charge: Rs " + charge[i]
            + "\n----------\n";
    }
    JOptionPane.showMessageDialog(null, msg);
}
static void issueBook() {
```

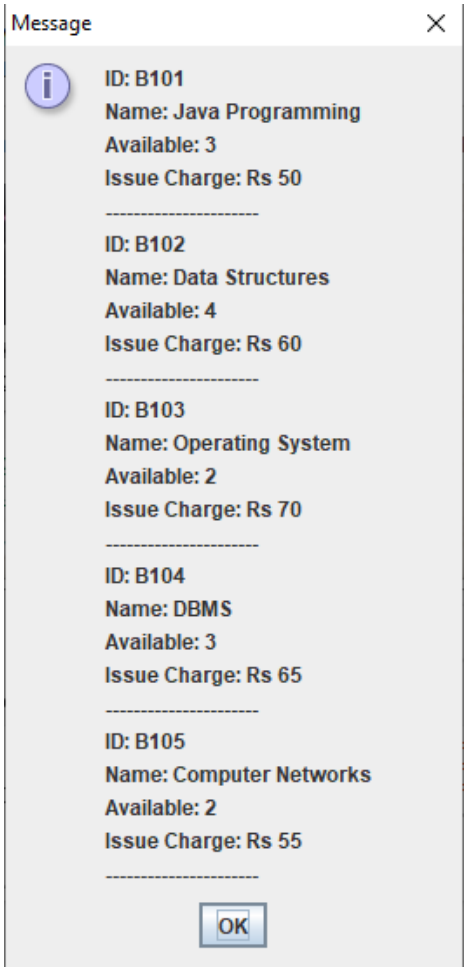
```
    if (!registered) {
        JOptionPane.showMessageDialog(null,
            "You are not registered!\nPlease make library card first.");
        return;
    }
    String id = JOptionPane.showInputDialog("Enter Book ID to Issue:");
    for (int i = 0; i < bookId.length; i++) {
        if (bookId[i].equalsIgnoreCase(id)) {
            if (quantity[i] > 0) {
                quantity[i]--;
                JOptionPane.showMessageDialog(null,
                    "Book Issued Successfully!\n"
                  + "Issue Charge: Rs " + charge[i]);
            } else {
                JOptionPane.showMessageDialog(null, "Book not available!");
            }
            return;
        }
    }
    JOptionPane.showMessageDialog(null, "Invalid Book ID!");
}
static void returnBook() {
    if (!registered) {
        JOptionPane.showMessageDialog(null, "Please register first!");
        return;
    }
    String id = JOptionPane.showInputDialog("Enter Book ID to Return:");
    int delivery = 30; // fixed delivery charge
```

```
        for (int i = 0; i < bookId.length; i++) {
            if (bookId[i].equalsIgnoreCase(id)) {
                quantity[i]++;
                int total = charge[i] + delivery;
                String receipt =
                    "======= BILL RECEIPT =======\n\n"
                + "Student Name: " + name + "\n"
                + "Roll No: " + roll + "\n"
                + "Department: " + dept + "\n\n"
                + "Book ID: " + bookId[i] + "\n"
                + "Book Name: " + bookName[i] + "\n\n"
                + "Issue Charge: Rs " + charge[i] + "\n"
                + "Delivery Charges: Rs " + delivery + "\n"
                + "--------------------------\n"
                + "TOTAL BILL: Rs " + total + "\n\n"
                + "Thank You!";
                JOptionPane.showMessageDialog(null, receipt);
                return;
            }
        }
        JOptionPane.showMessageDialog(null, "Invalid Book ID!");
    }
}
```

Output


Input
LIBRARY MANAGEMENT SYSTEM
1. Make Library Card
2. View Library Card
3. View Books
4. Issue Book
5. Return Book (Bill Receipt)
6. Exit
Enter choice:
OK
Cancel


Message
------ LIBRARY CARD ------
Name: ABDUL MOHID KASHIF
Roll No: 2024F-BCE-161
Department: COMPUTER ENGINEERING
Semester: 3
Phone: 3343397828
Email: MOHID@gmail.com
Address: A-12 Block 6 Gulshan e Iqbal Karachi
OK


Message
ID: B101
Name: Java Programming
Available: 3
Issue Charge: Rs 50

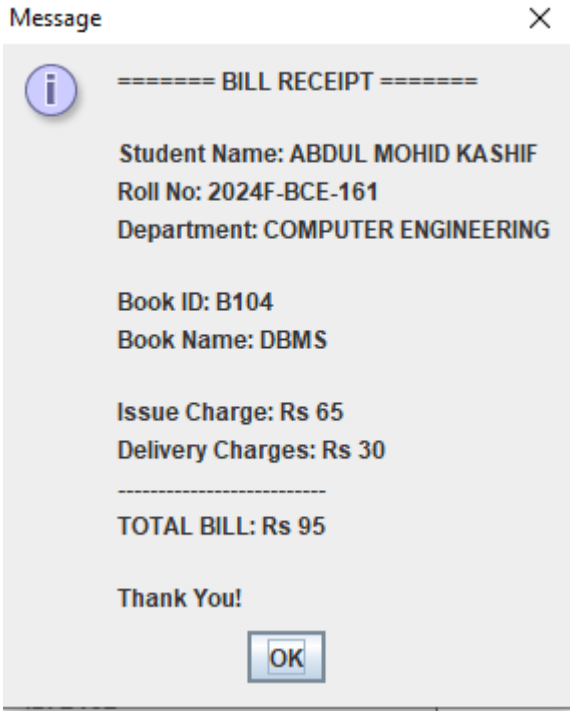
ID: B102
Name: Data Structures
Available: 4
Issue Charge: Rs 60

ID: B103
Name: Operating System
Available: 2
Issue Charge: Rs 70

ID: B104
Name: DBMS
Available: 3
Issue Charge: Rs 65

ID: B105
Name: Computer Networks
Available: 2
Issue Charge: Rs 55

OK


Message
======= BILL RECEIPT =======
Student Name: ABDUL MOHID KASHIF
Roll No: 2024F-BCE-161
Department: COMPUTER ENGINEERING
Book ID: B104
Book Name: DBMS
Issue Charge: Rs 65
Delivery Charges: Rs 30

TOTAL BILL: Rs 95
Thank You!
OK

14. Conclusion and Recommendations

The Library Management System successfully automates basic library operations. It is easy to use, cost-effective, and suitable for beginner-level projects. Future improvements may include database integration and advanced reporting.

15. Acknowledgements

I would like to thank my teacher and institution for their guidance and support during the development of this project.

16. References

- Java Documentation
- NetBeans IDE User Guide
- Lab File